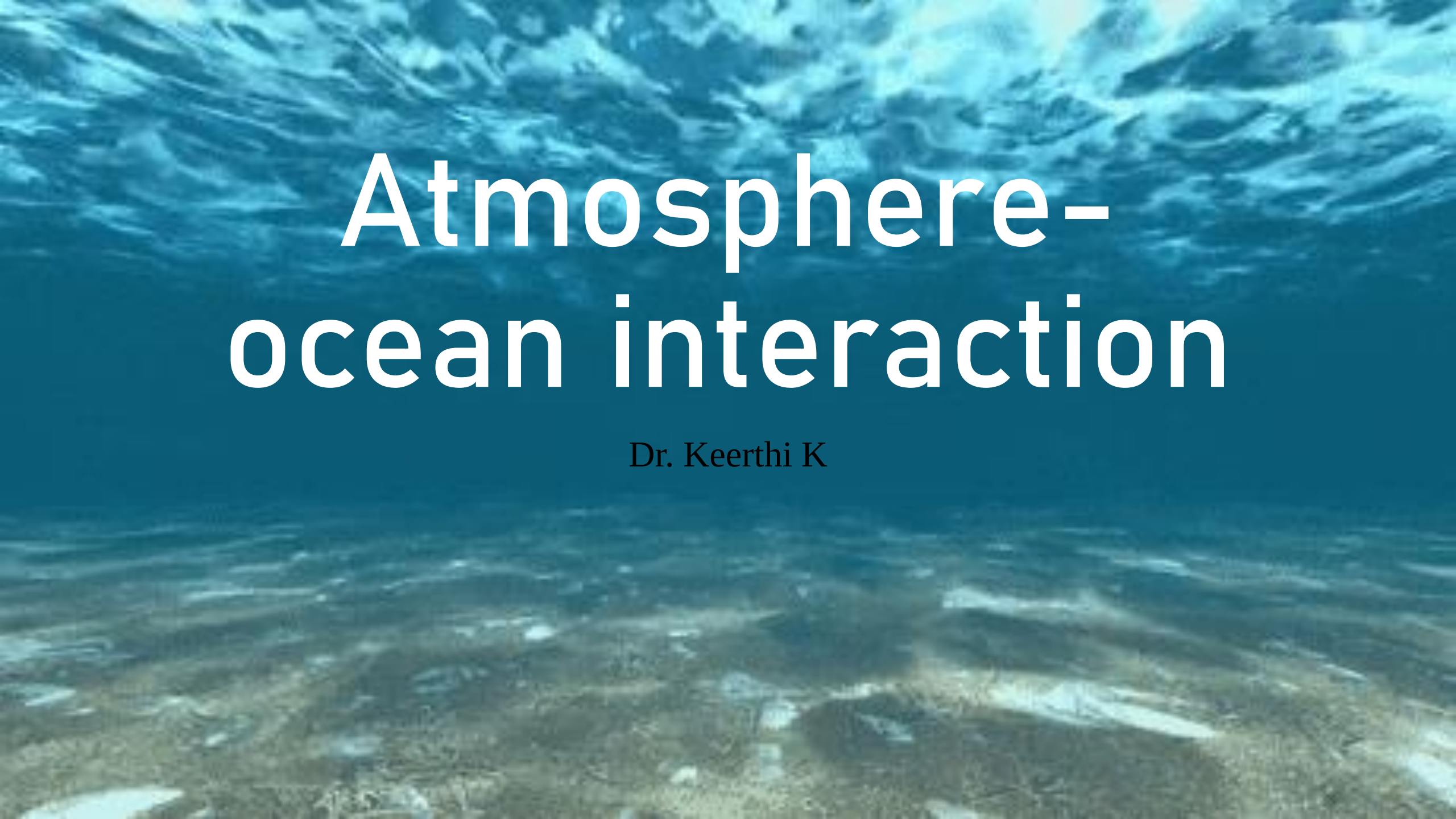
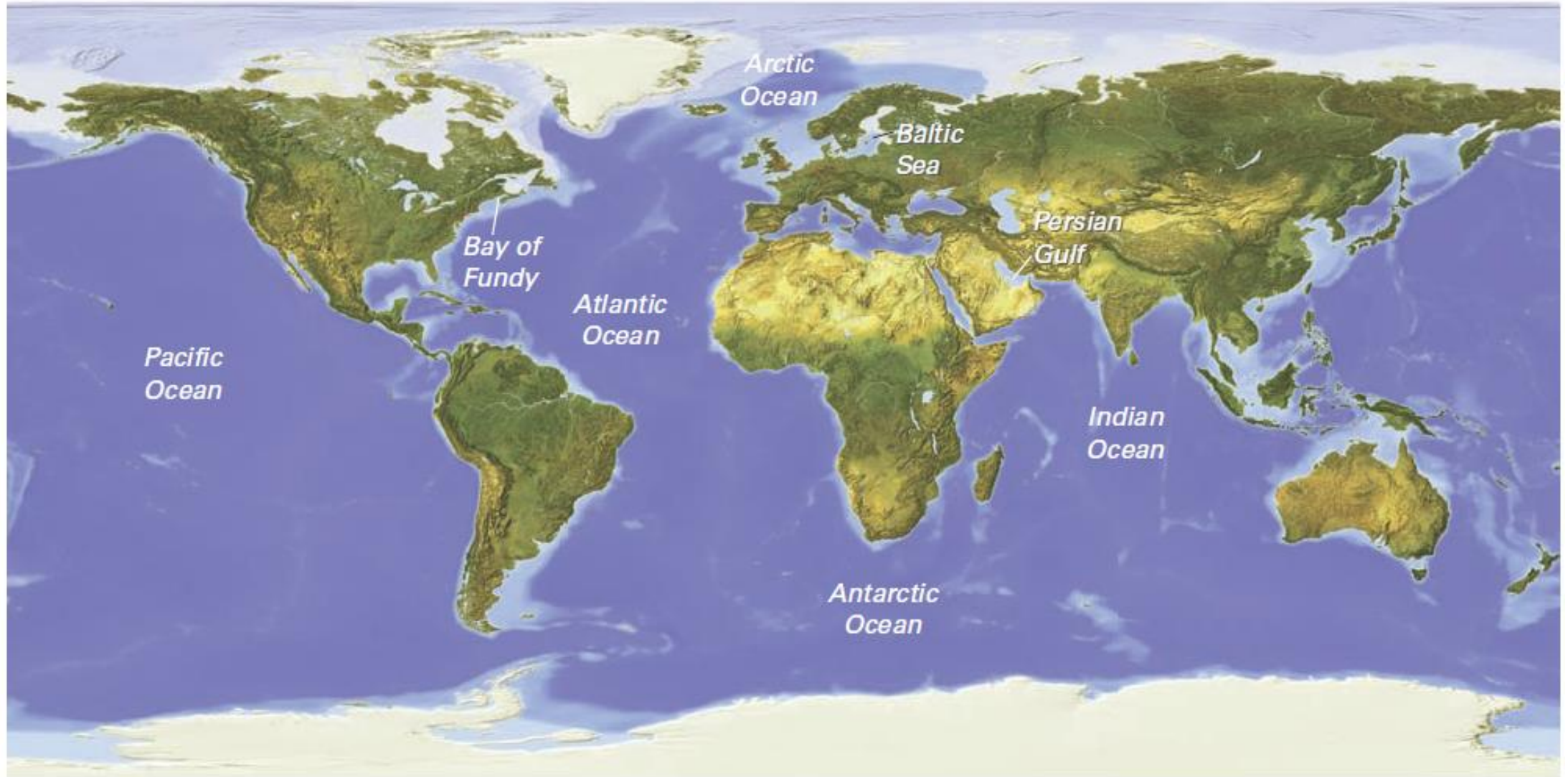




Atmosphere- ocean interaction

Dr. Keerthi K

Ocean Geography



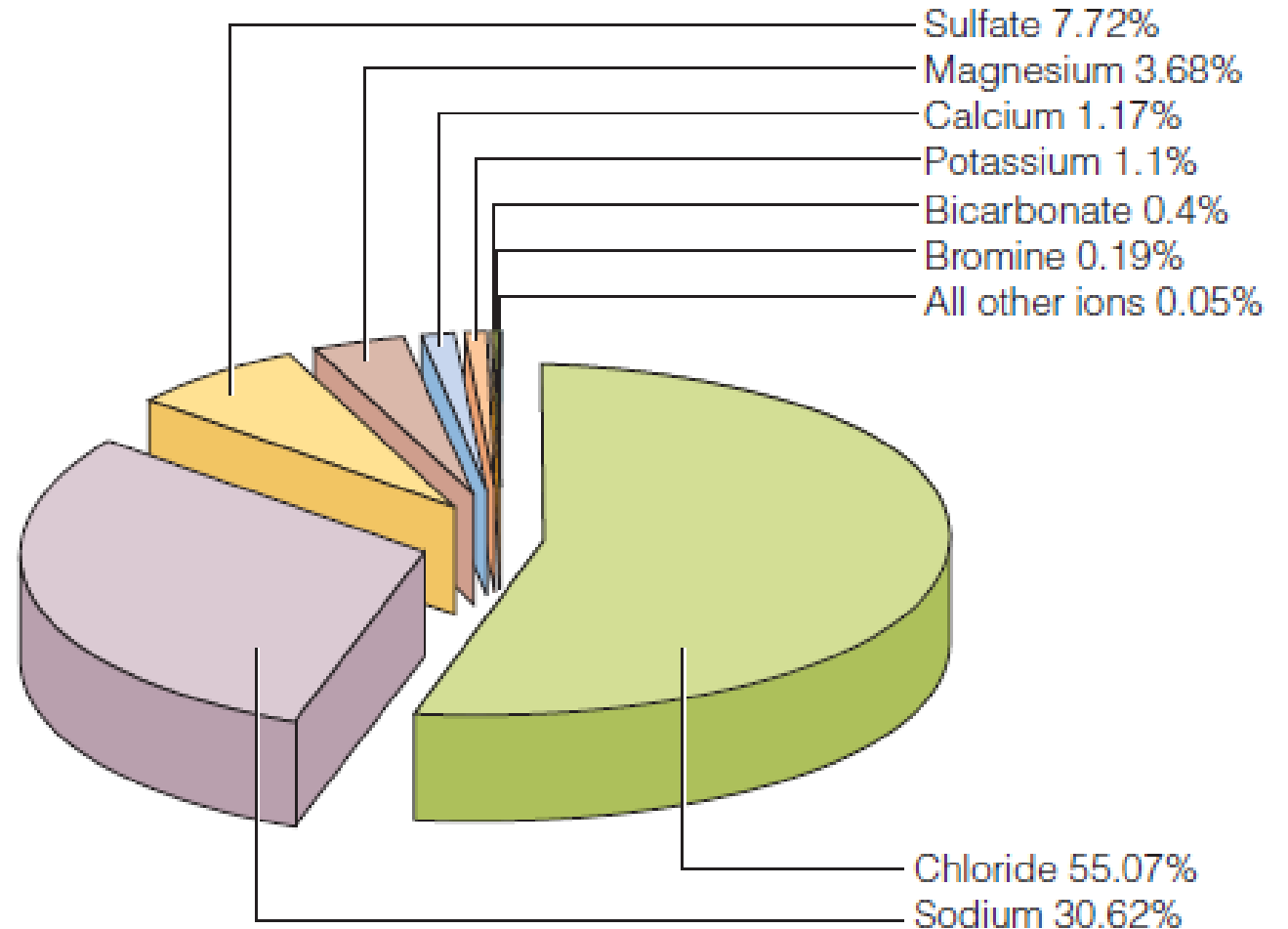
Water chemistry

Salts and Trace Elements

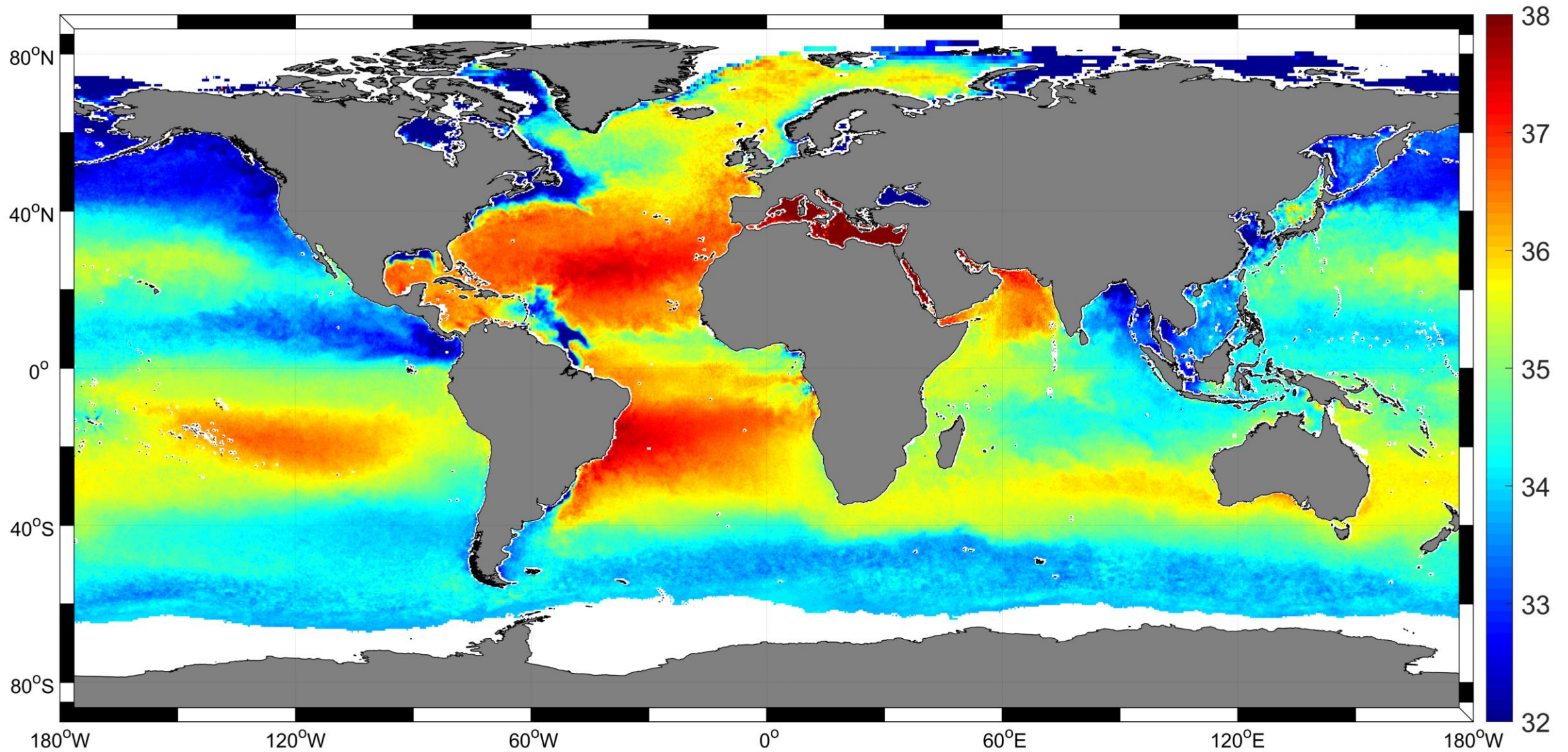
- The **salinity** of seawater is the total quantity of dissolved salts, expressed as a percentage.

Dissolved Gases

- In addition to trace elements and salts, seawater also contains dissolved gases, especially carbon dioxide and oxygen.



Ocean salinity

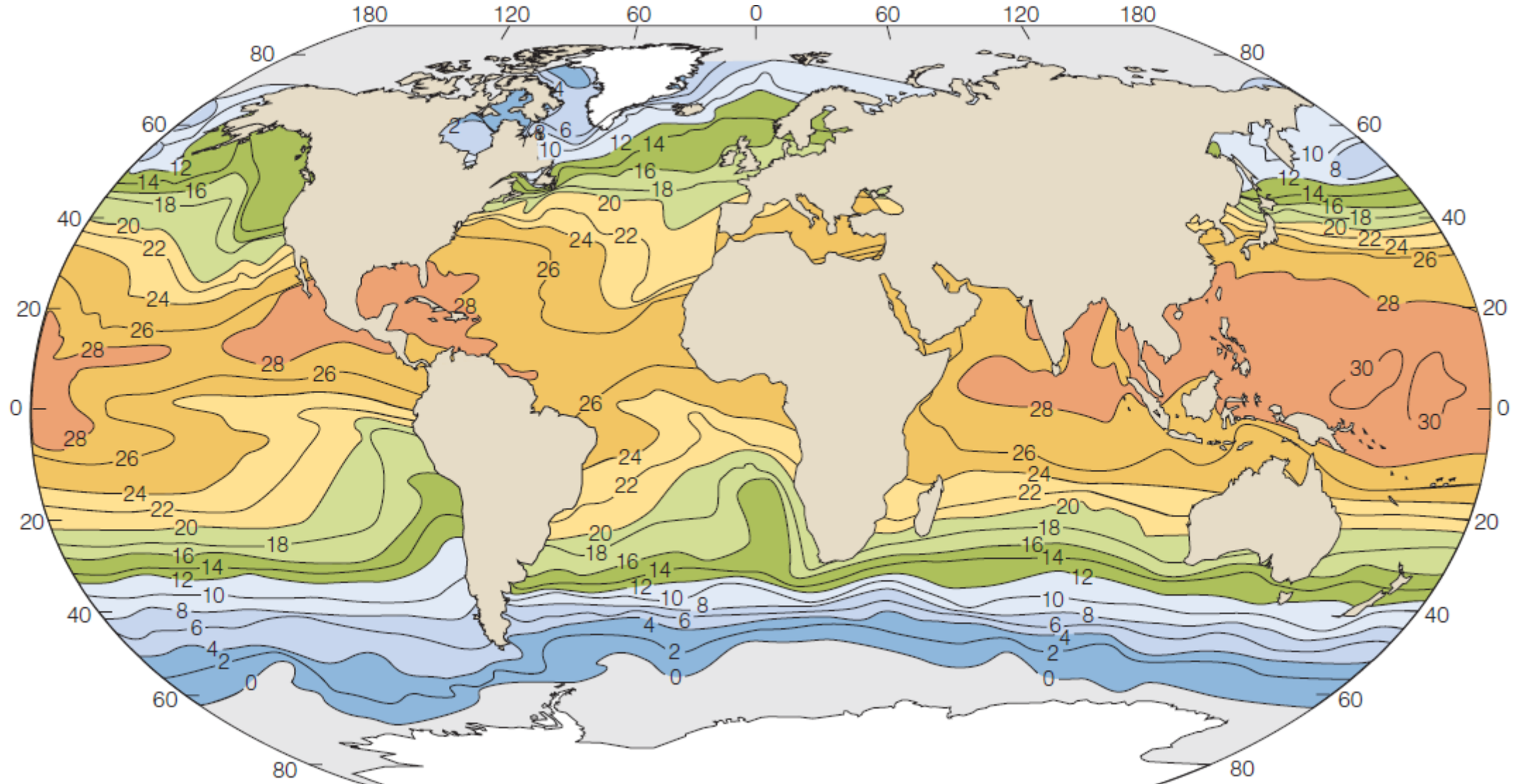


Factors that control salinity

The most important factors affecting salinity are:

1. **Evaporation**, which removes fresh water and leaves the remaining water saltier;
2. **Precipitation** of rain and snow, which adds fresh water, thereby diluting the seawater and making it less salty;
3. **Inflow of fresh (river) water**, which makes the seawater less salty; and
4. **Freezing of sea ice** because when seawater freezes salts are excluded from the ice, leaving the unfrozen seawater saltier.

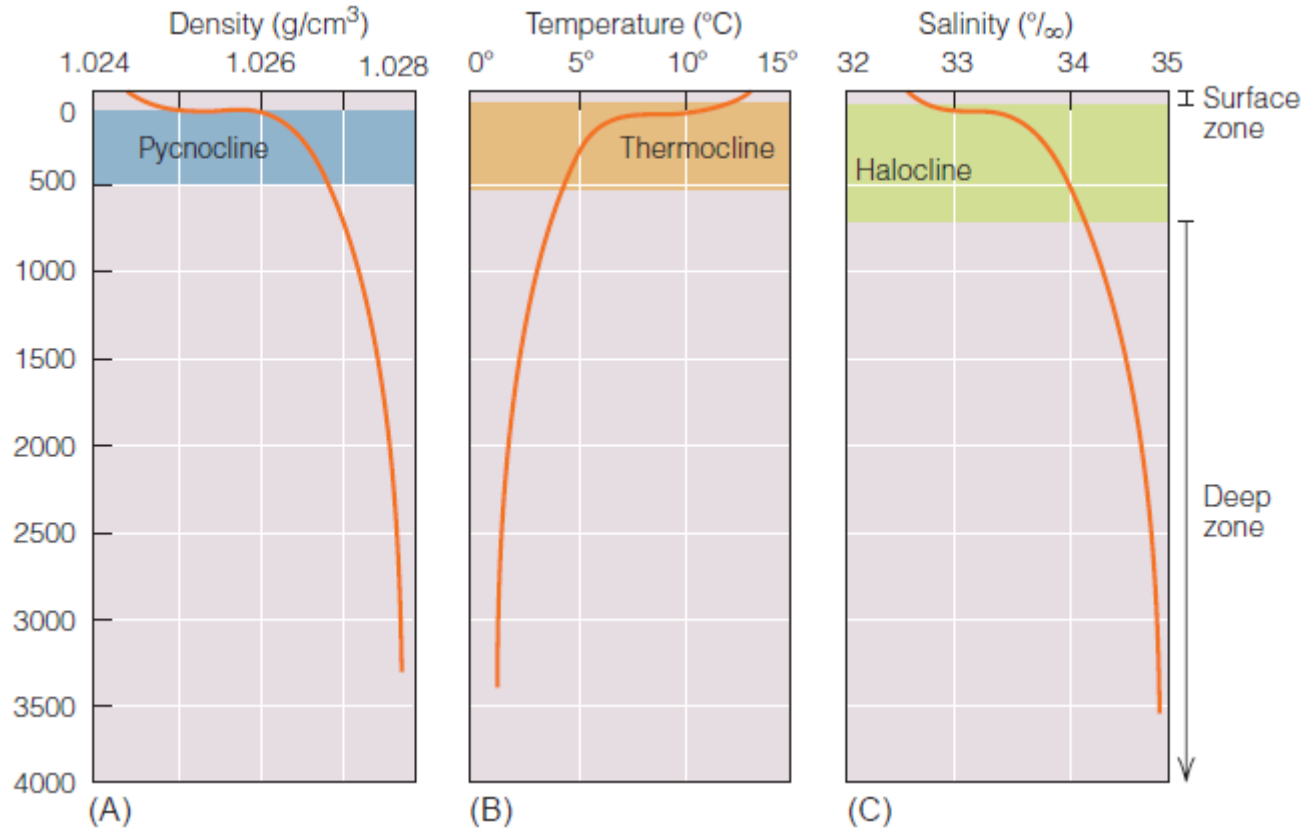
Sea-surface temperatures



This map shows typical sea-surface temperatures (in °C) in the world ocean during August

Vertical Stratification of the Ocean

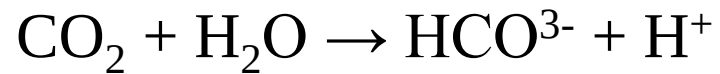
- Seawater density depends on temperature and salinity
 - Hot and fresh water is least dense.
 - Cold and salt water is most dense
- Oceanographers recognize three major depth zones in the ocean.
- The *surface zone*, typically extending to a depth of 100 to 500 m, consists of relatively warm water (except in polar latitudes, where the surface zone is absent).



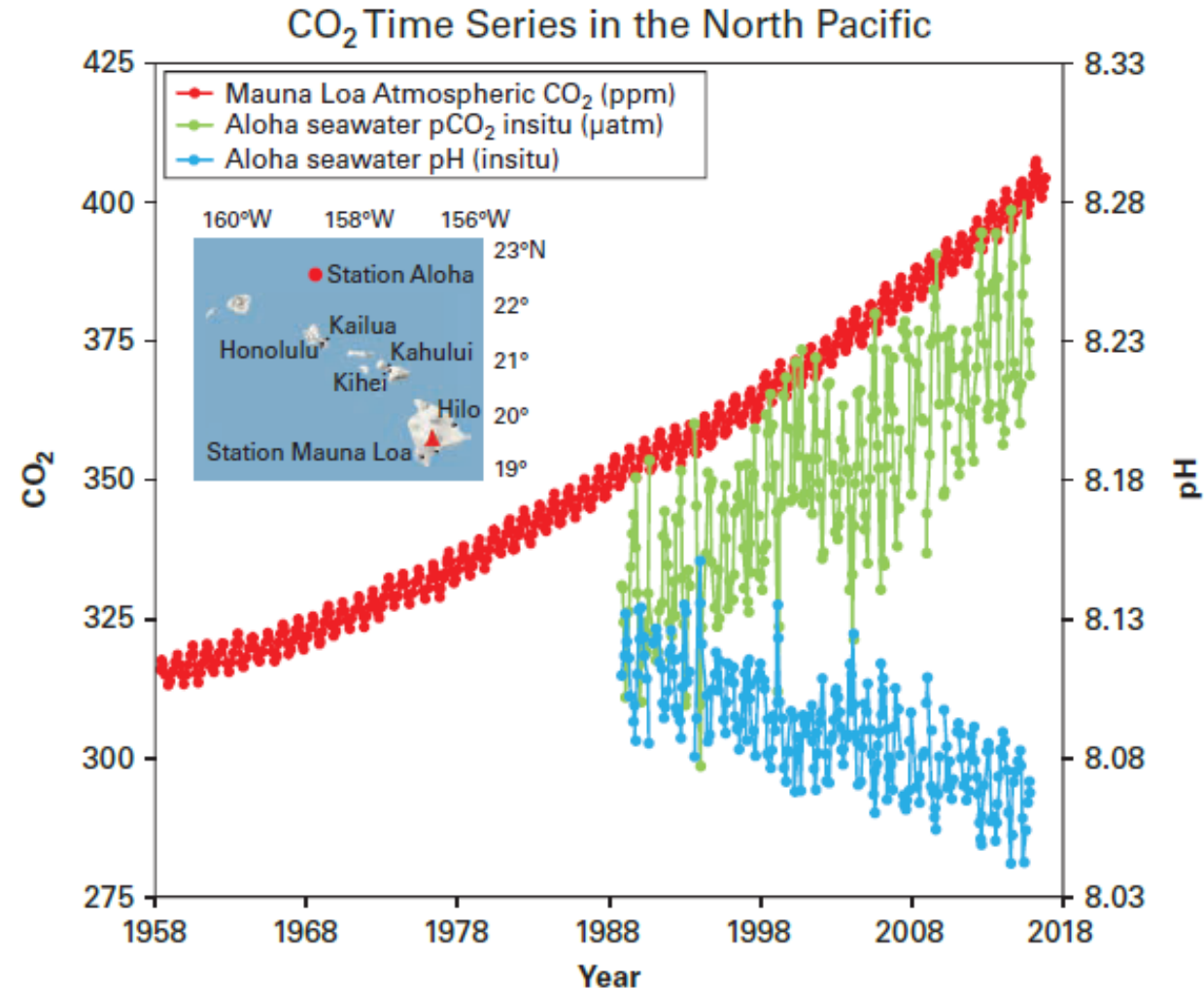
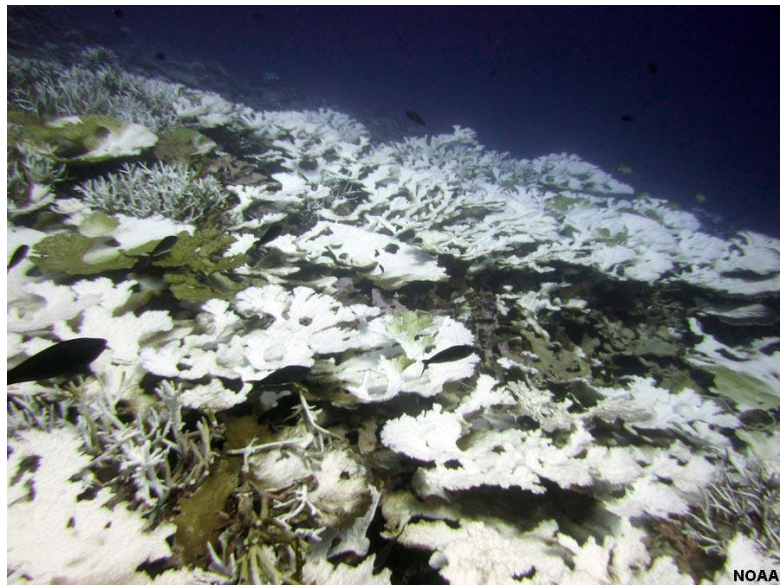
pycnocline (A), a zone in which density increases with depth;
thermocline (B), a zone in which temperature decreases with depth;
halocline (C), a zone in which salinity increases with depth.

Ocean acidification

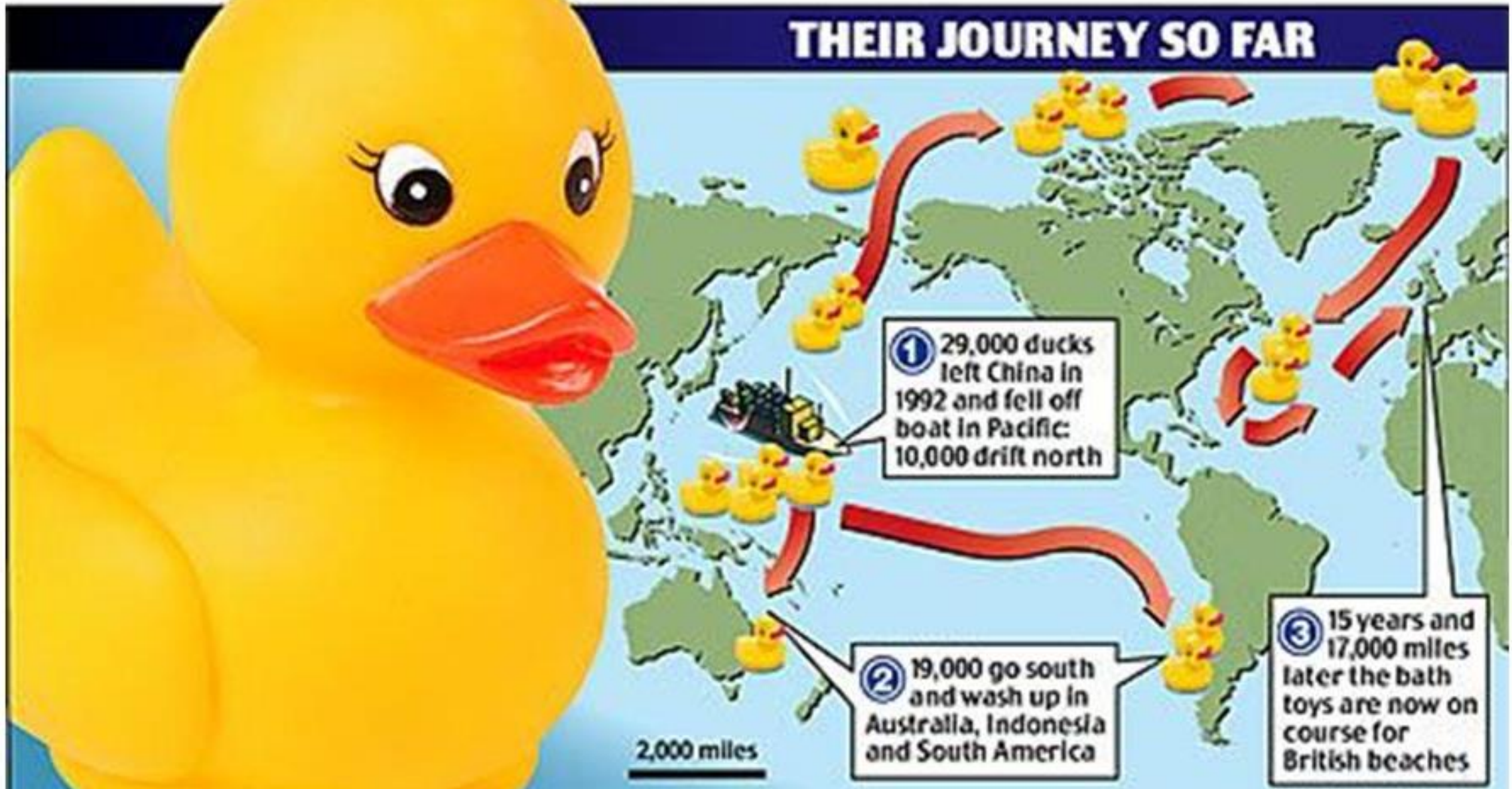
- As more CO₂ from the atmosphere diffuses into the oceans and converts to carbonic acid, the pH of the ocean as a whole drops. This process is called **ocean acidification**.



- The organisms try to make hard shells (CaCO₃) but it dissolves in acid – so they spend their energy to make shells and eventually die leaving white shells behind.



Ocean circulation



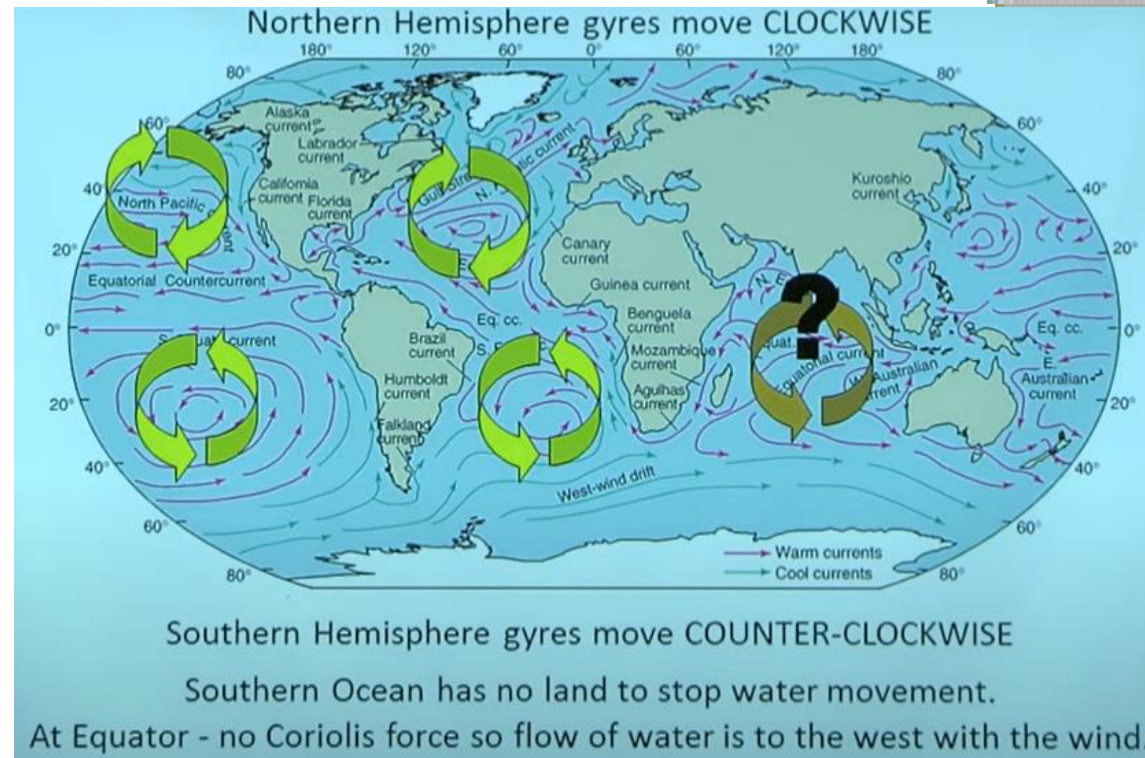
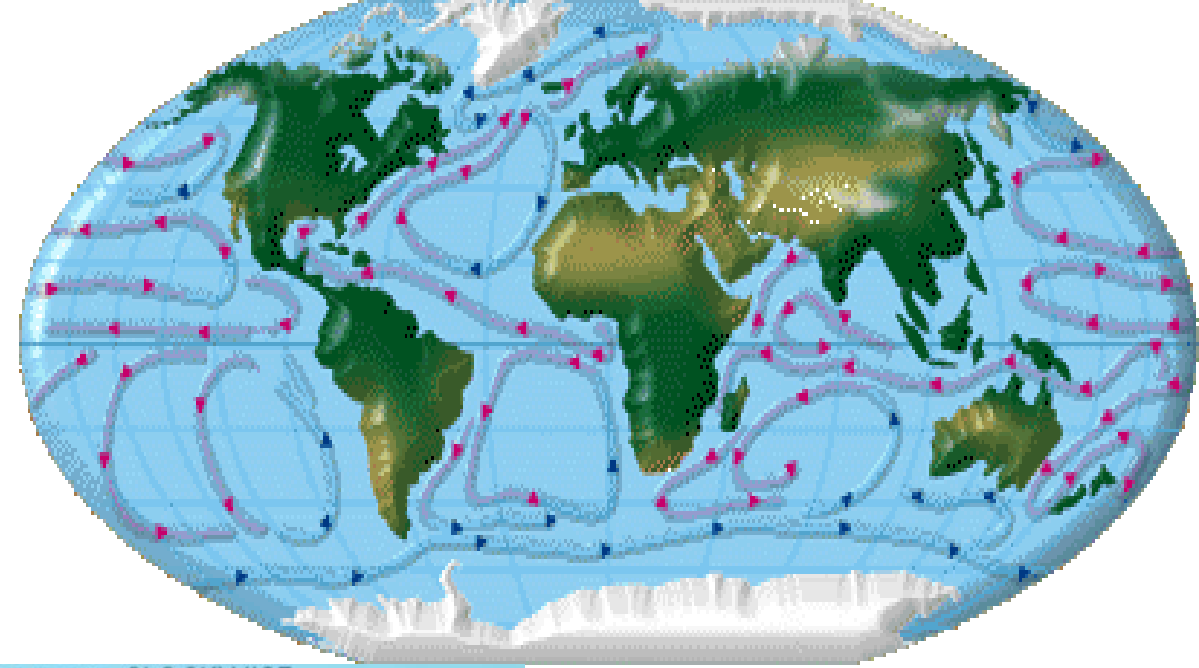
Ocean currents

- A **current** is a continuous flow of water in a particular direction. In general, **ocean currents follow prevailing wind directions except where the current encounters an obstacle, such as a landmass.**
- Currents in the oceans include surface currents and deep currents:
 - **Surface currents** are driven horizontally by **effects of the wind (upper 100 m – 400 m).**
 - **Deep currents** are driven horizontally and vertically by **differences in density** (density changes typically start near the surface).
- Ocean circulation is also influenced by seawater temperature and density.
 - Warm water in the tropics flows in currents to polar regions where it cools and the formation of sea ice concentrates the salt in seawater, increasing its density so that it sinks.
 - Cold and salty water (concentrated by surface evaporation) sinks. Elsewhere seawater rises where it is displaced by colder and saltier water.



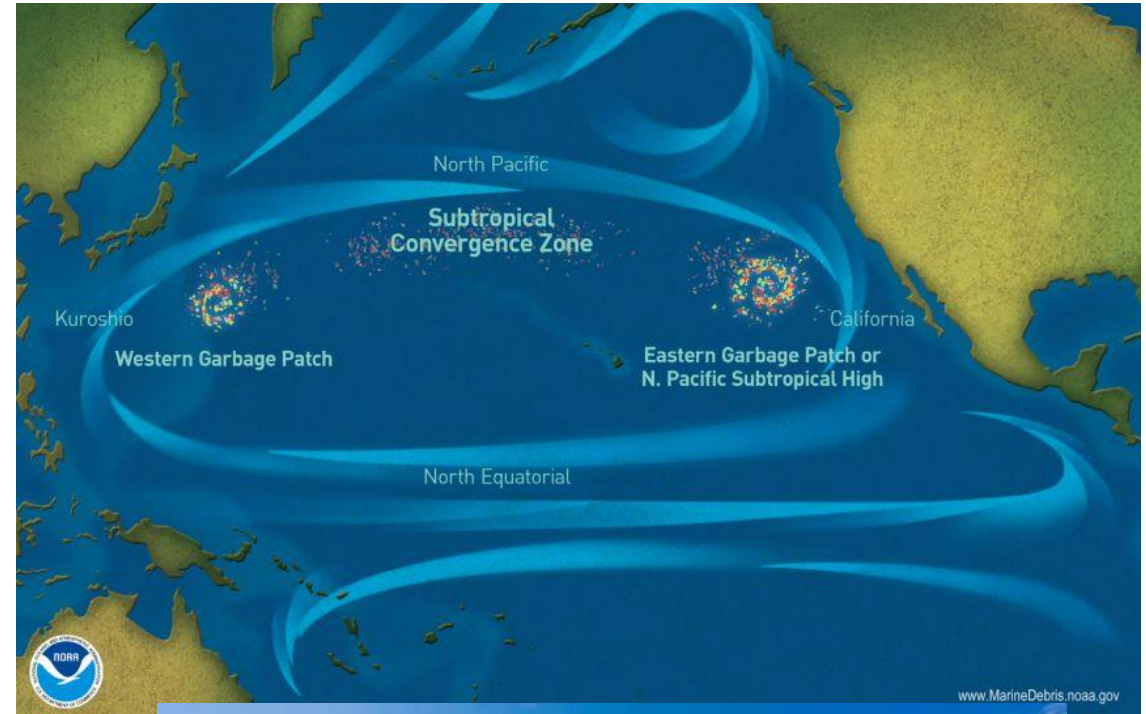
Ocean - Surface currents

- Surface currents are driven primarily by friction between wind blowing over the sea surface and surface water.
- The winds simply drag the sea surface along in the same direction that the winds are blowing.
- On a global scale, **circulation patterns in the atmosphere generate near-circular patterns of ocean currents known as *gyres*.**



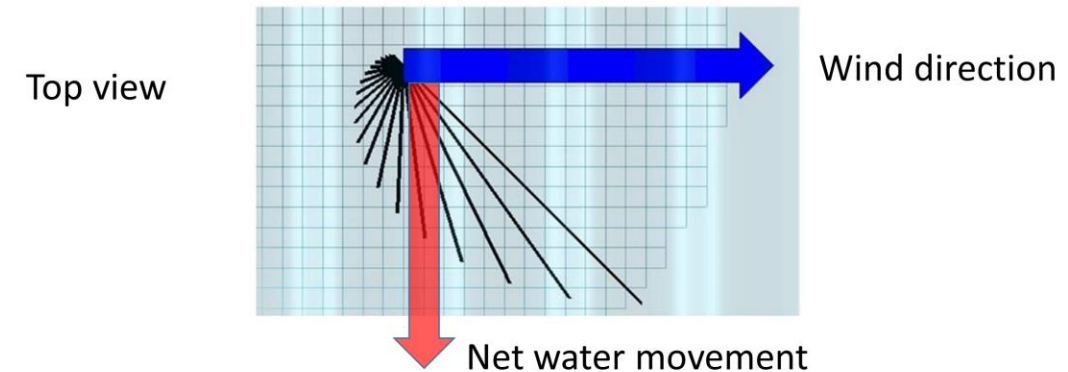
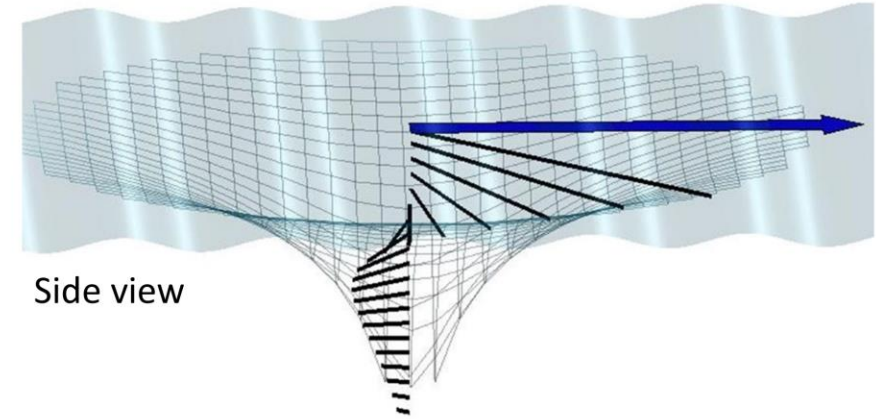
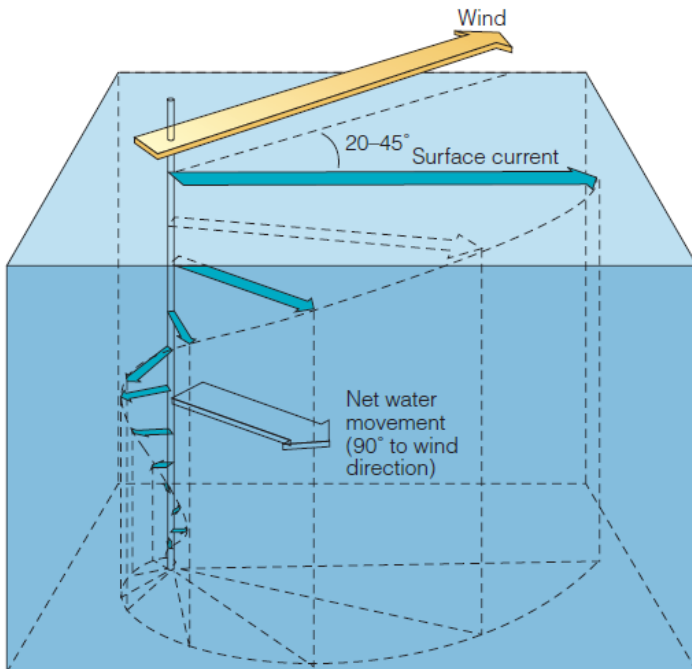
What is a Garbage Patch?

- A **garbage patch** is a popular name for concentrations of marine debris (mostly small pieces of floating plastic) that accumulate across the more stagnant central parts of the large gyres in the ocean basins.
- The central regions of ocean basins are areas of convergence and downwelling, so trash from sources on land and sea are carried long distances by currents, much of it ending up in a convergence zone garbage patch.
- The largest garbage patch is in the north Pacific Ocean.

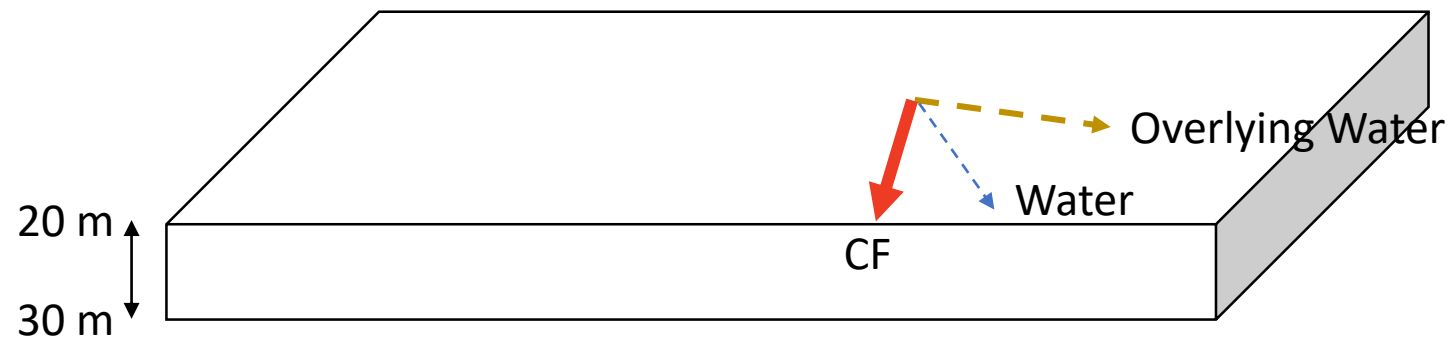
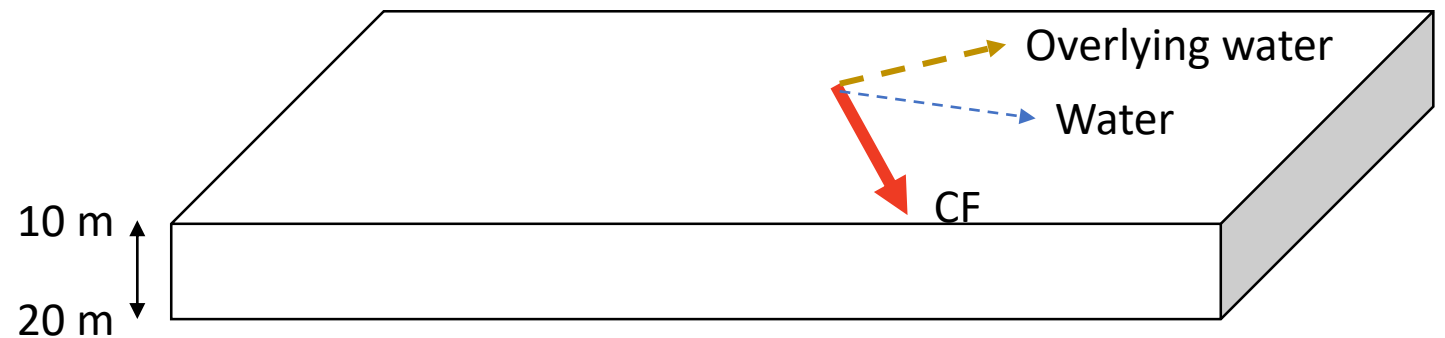
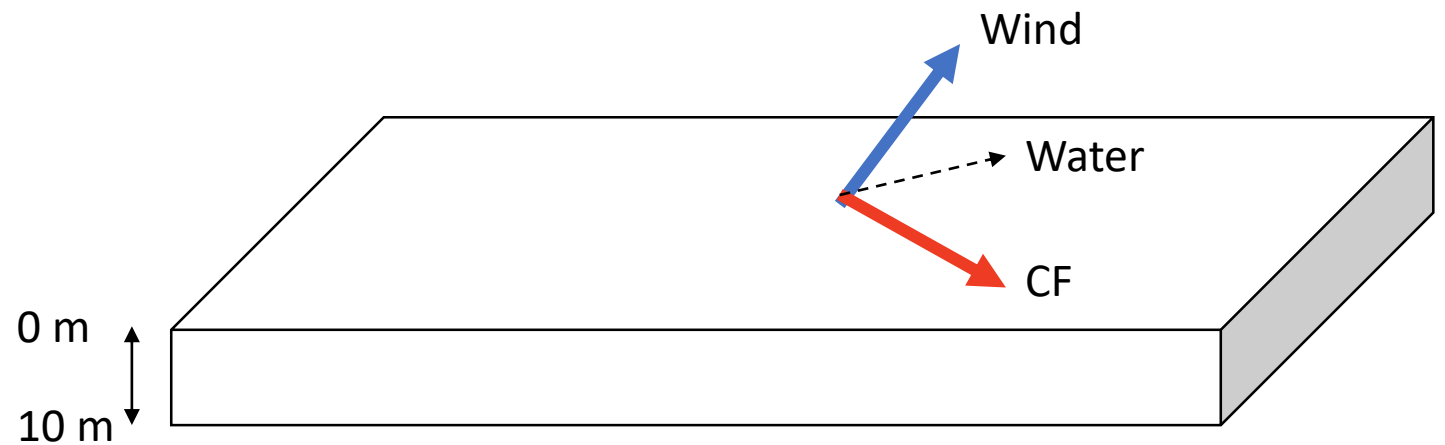


Surface currents - Ekman Transport

- Wind drags water.
- **Ekman spiral** – looking vertically down at the current they seem to spiral.
- **Ekman transport** – average flow of water throughout the spiral is $\sim 90^\circ$ to the right/left of the surface wind in the Northern/southern hemisphere.

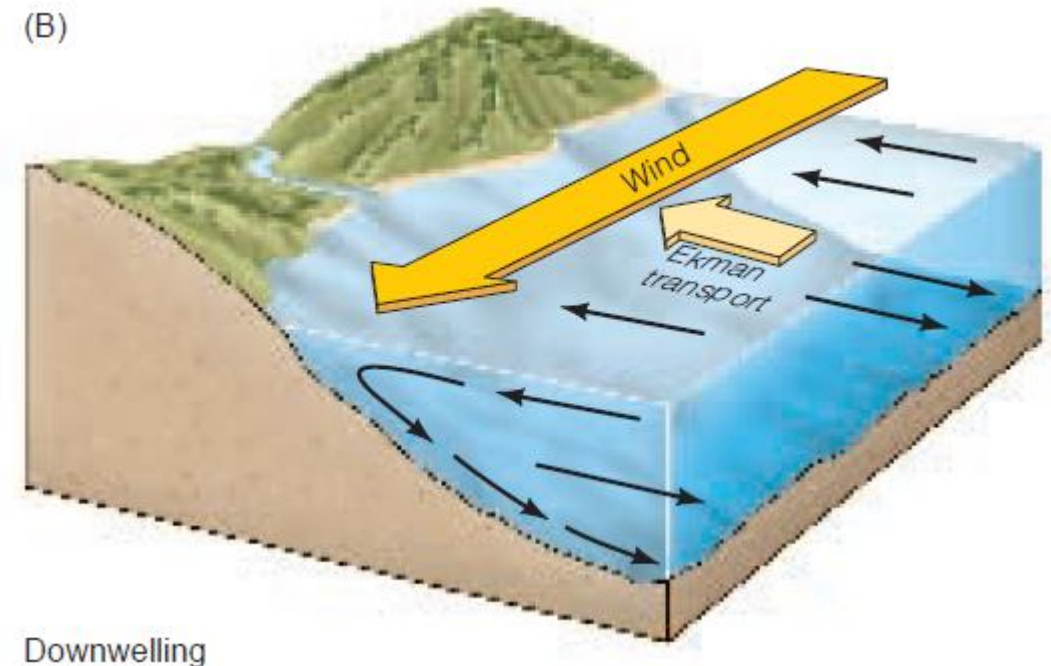
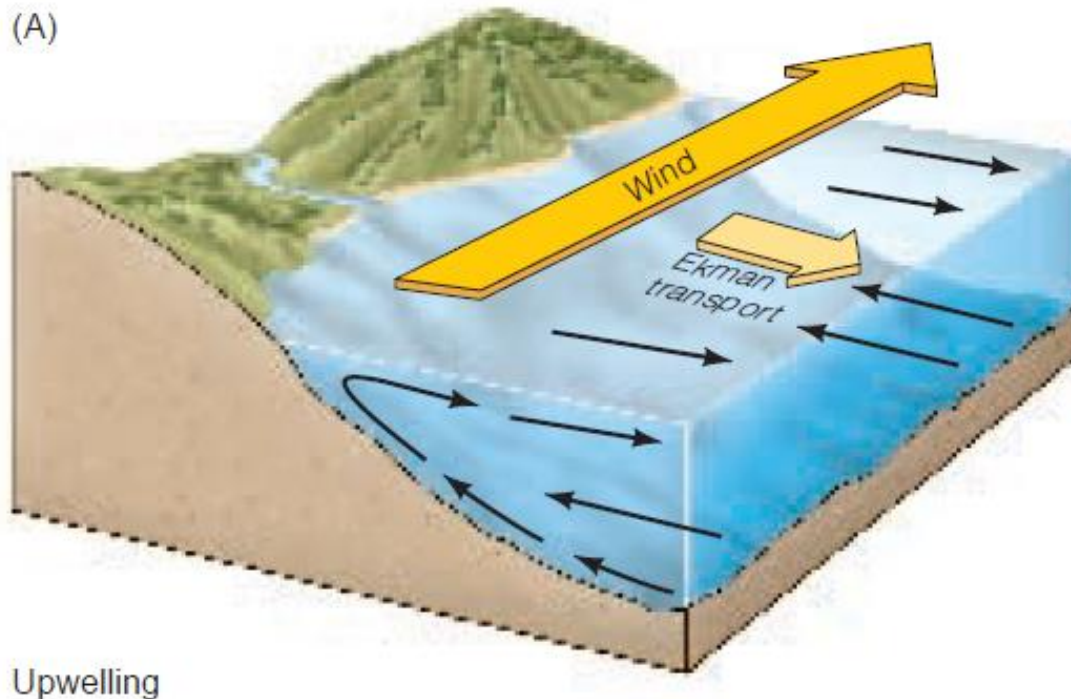


- Winds exert a force on the surface.
- Friction translates this force down into the water column.
- Without Coriolis, surface water would flow with the applied winds. Coriolis force acts 90° to the direction of movement so surface water actually flows at 20 - 45°
- Water beneath the layer feels friction/drag from the layer moving above.
- Coriolis force again acts at 90° to movement so this layer moves at even more of an angle compared to wind.
- Process repeats – layers look like spiral – net direction is 90° to the wind.



Surface currents: Upwelling/downwelling

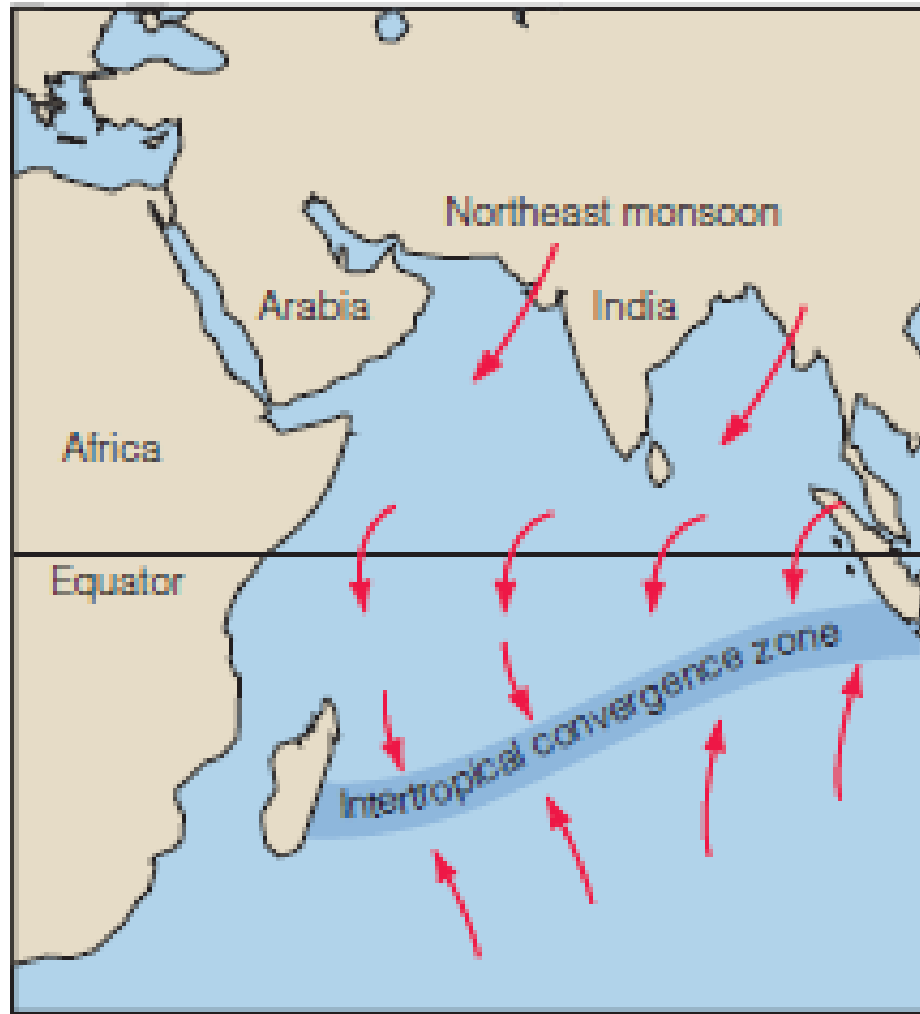
- **Upwelling** occurs when winds cause surface waters to be transported away from the coast, allowing colder, nutrient-rich deep water to rise and replace the displaced water. This is crucial for marine productivity, as it supports the growth of phytoplankton.
- **Downwelling** occurs when Ekman transport pushes surface waters toward the coast, forcing surface water to sink. This brings oxygenated water to deeper layers.



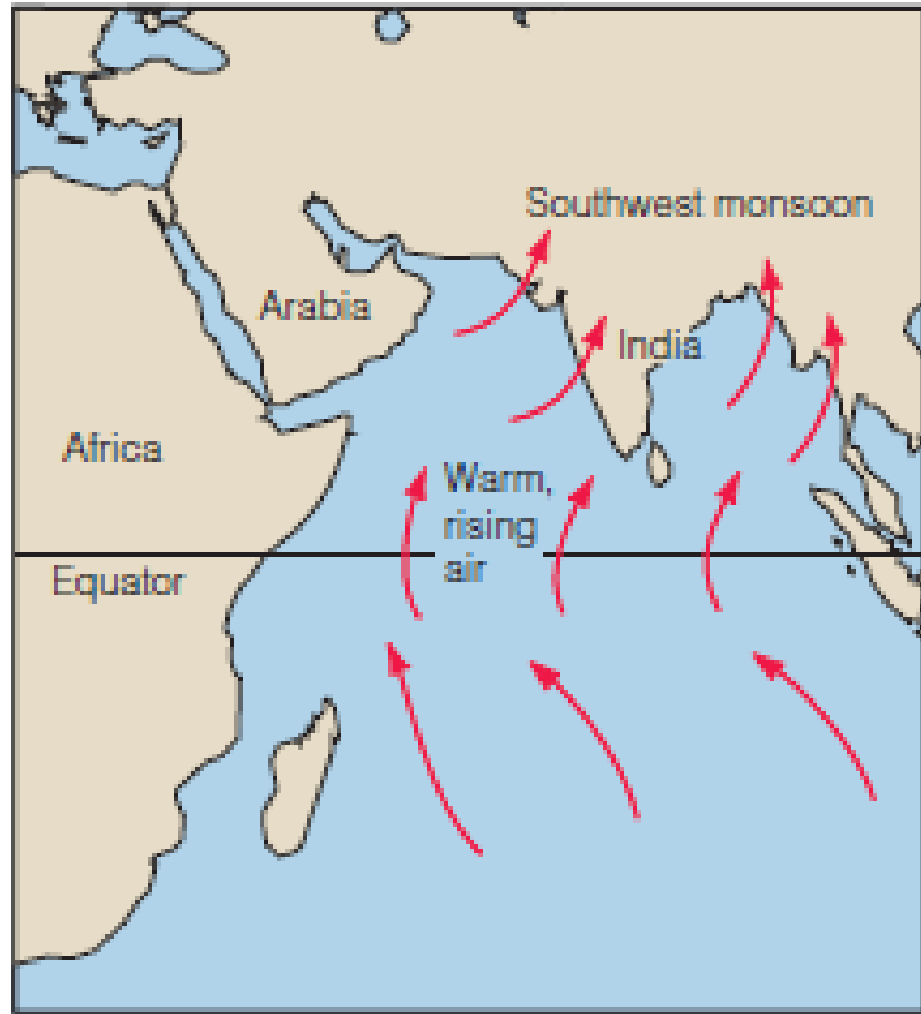


**Wind
direction**

(A) Winter

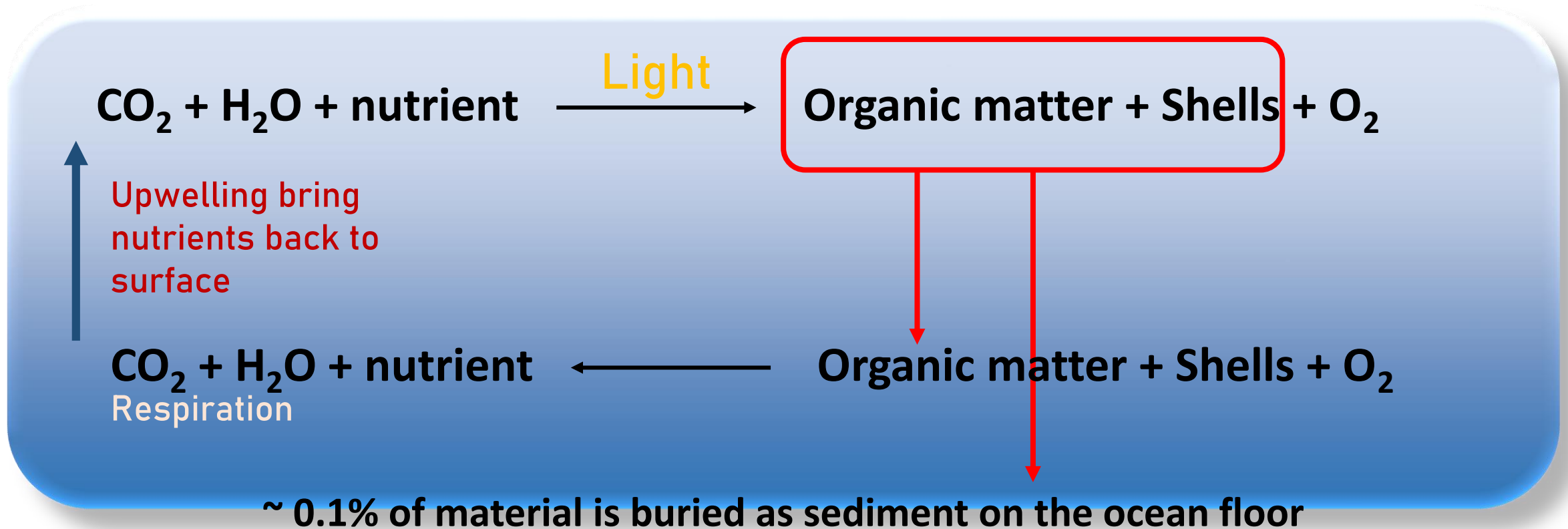


(B) Summer

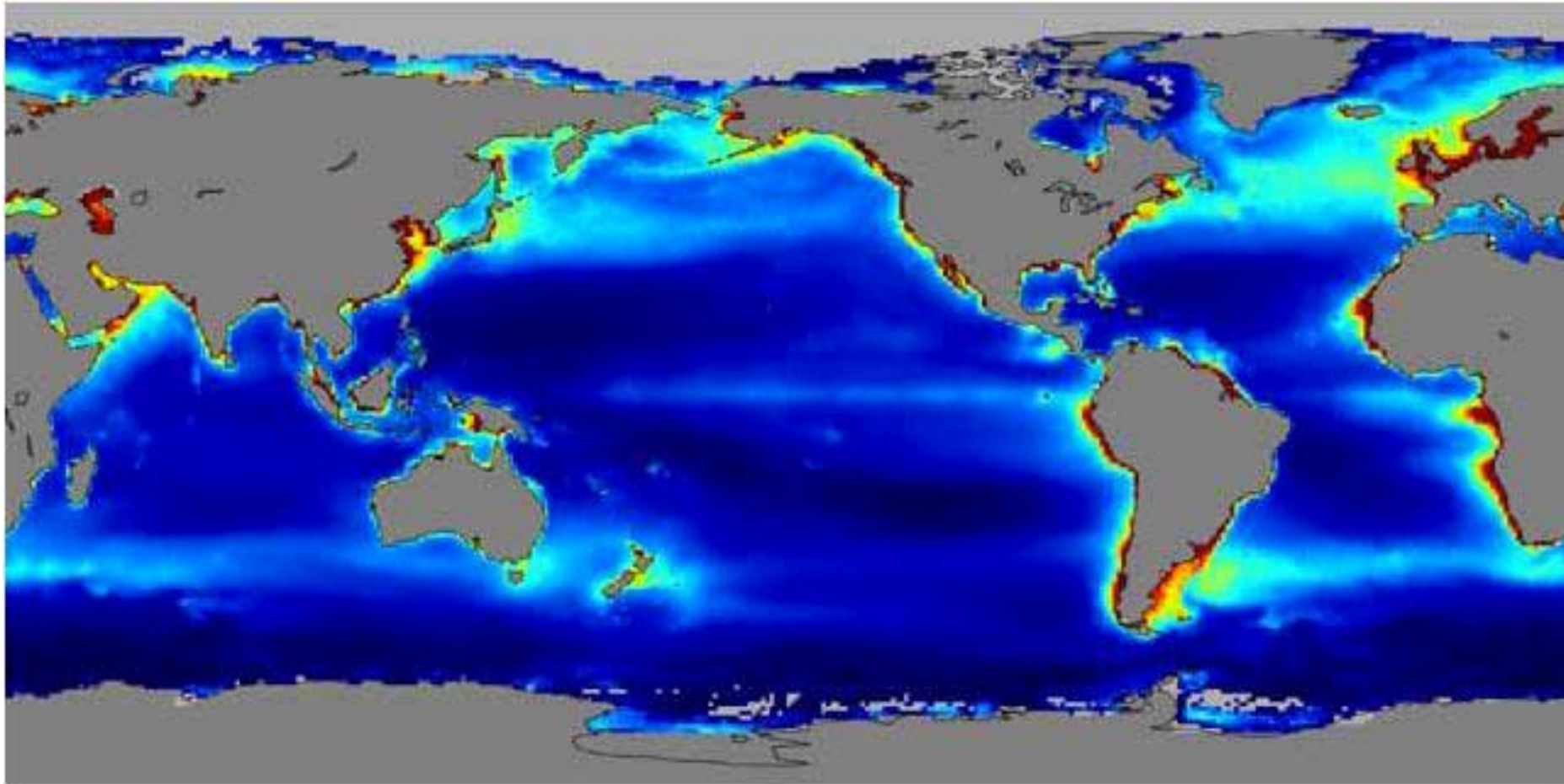


Oceans: Life

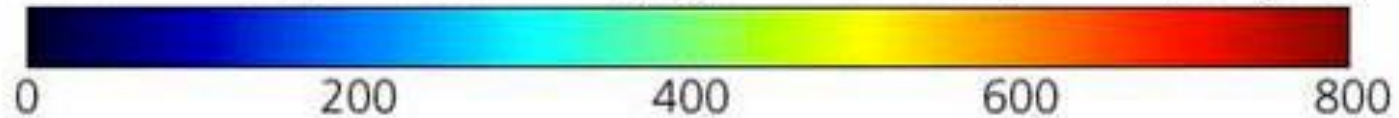
- Huge variety of life in the oceans – from tiny bacteria to blue whales
- Most important type are the small floating organisms (plankton).
- Some plankton in surface ocean photosynthesize to produce organic matter.
- Some also produce hard shells of calcium carbonate (CaCO_3) or silica (SiO_2)



Global Distribution of life



Net Primary Productivity (grams Carbon per m² per year)

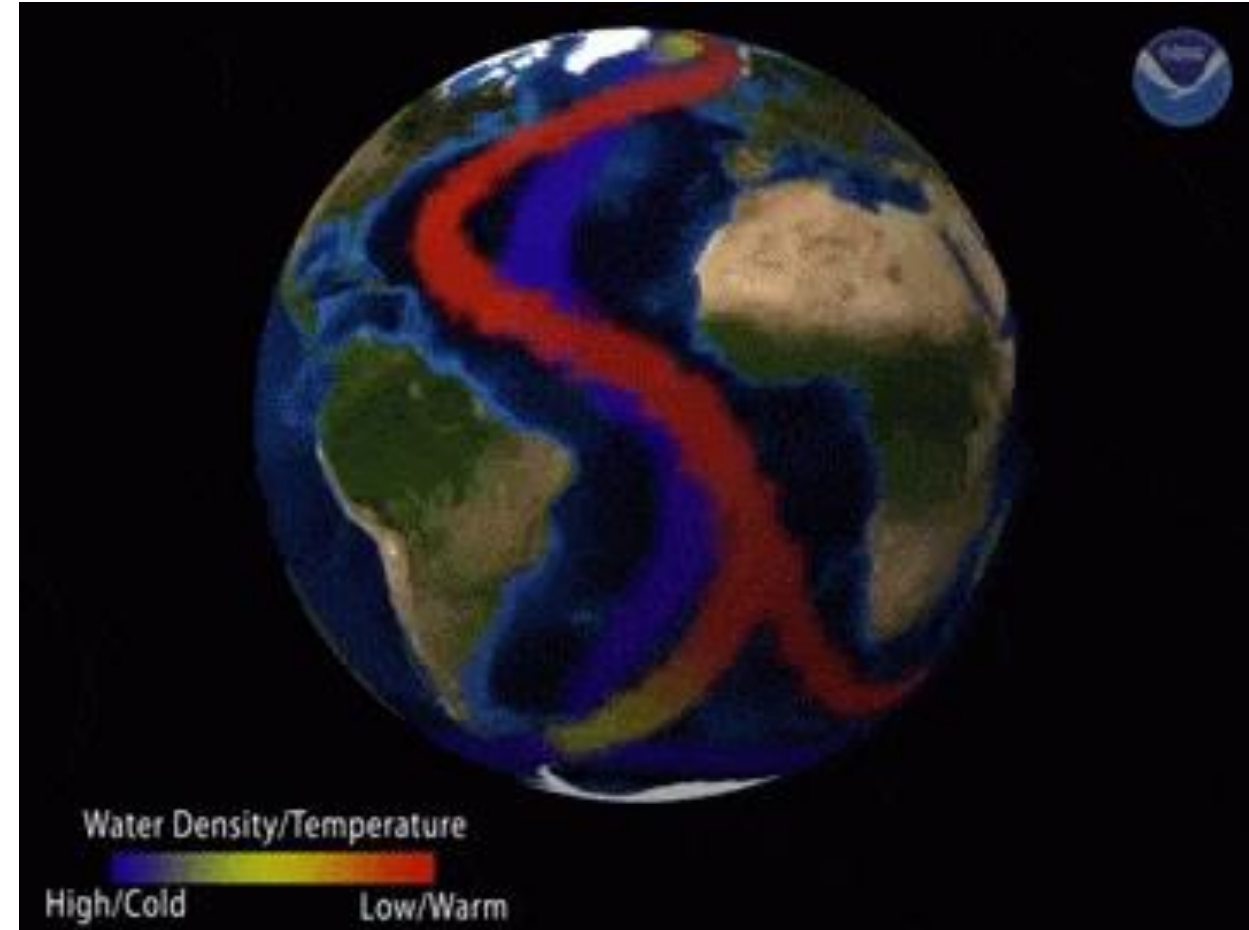
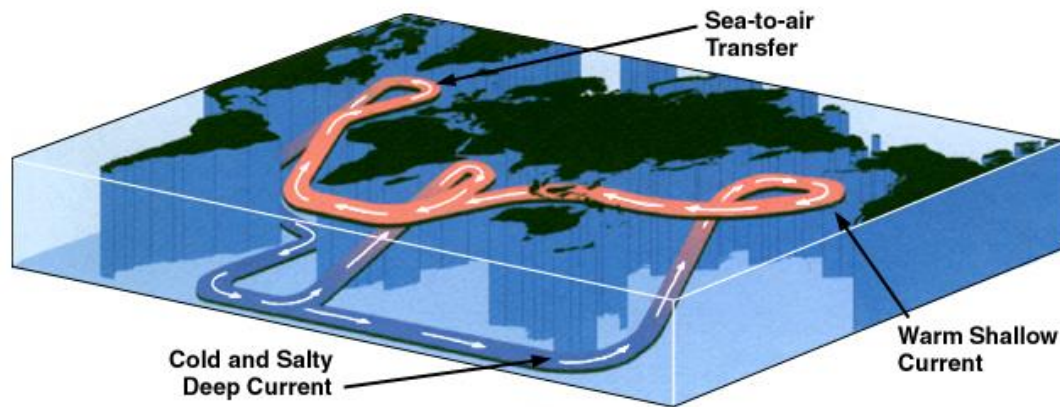


Ocean circulation: Deep currents

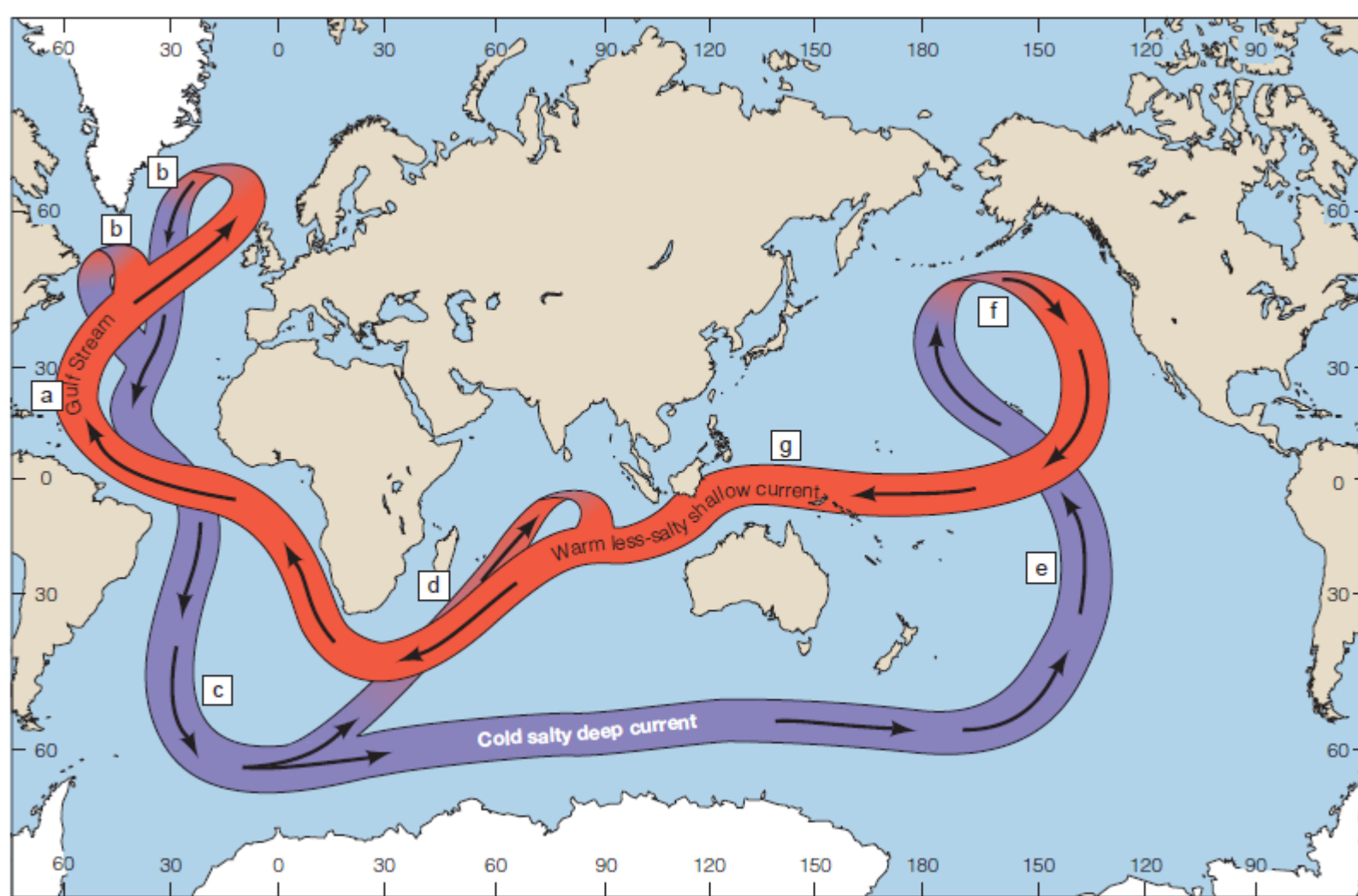
Thermohaline Circulation

This deep-water current is known as the **global conveyor belt**.

- Thermohaline circulation is driven by density not winds.
- Thermo = temperature; Haline = salinity
- Coldest and saltiest water is the most dense.



Thermohaline Circulation



Gulf Stream (*a*)
North Atlantic (*b*)
South Atlantic (*c*),
Indian (*d*) and Pacific (*e*) oceans.
Surface water warmed by solar energy
flows into the western Pacific and Atlantic
basins (*g*) to close the loop of the great
global thermohaline cells.

Thermohaline Circulation

- The global conveyor belt moves much more slowly than surface currents -- a few centimeters per second, compared to tens or hundreds of centimeters per second.
- Scientists estimate that it takes one section of the belt 1,000 years to complete one full circuit of the globe.
- However slow it is, though, it moves a vast amount of water -- more than 100 times the flow of the Amazon River.
- The global conveyor belt is crucial to the base of the world's food chain.
- As it transports water around the globe, it enriches carbon dioxide-poor, nutrient-depleted surface waters by carrying them through the ocean's deeper layers where those elements are abundant.
- The nutrients and carbon dioxide from the bottom layers that are distributed through the upper layers enable the growth of algae and seaweed that ultimately support all forms of life.
- The belt also helps to regulate temperatures.

What could cause the thermohaline circulation to slow or stop ?

